Tokenization Alignment for Stable Multi-Turn Reinforcement Learning in Large Language Models

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1 Introduction

Reinforcement learning from human feedback (RLHF) and its variants have emerged as the dominant paradigm for aligning large language models with desired behaviors (Ouyang et al., 2022). More recently, agent RLwhere a language model learns through multi-turn interaction with an environment or userhas attracted significant attention as a path toward capable autonomous systems. However, multi-turn RL introduces training dynamics that are considerably more complex than single-turn settings, and several sources of instability remain poorly understood.

One such source is *tokenization drift* (Li et al., 2025; The Agent Lightning , AGL): the mismatch between the token IDs sampled by the model during rollout and the token IDs produced when the same text is re-encoded during the training update. Because RL objectives such as PPO and GRPO operate on log-probabilities computed over token sequences, any divergence between the rollout token sequence and the training token sequence produces spurious off-policy updates that cannot be corrected through standard importance sampling, as the mismatch is not at the token level (The Agent Lightning , AGL).

In this proposal, we identify an additional, structurally distinct cause of tokenization drift that persists even when assistant token IDs are faithfully preserved. Specifically, in multi-turn RL, the full training sequence is constructed by concatenating system prompt tokens, user turn tokens (encoded from text), and assistant turn tokens (preserved from generation). When overlapping vocabulary items appear in both user and assistant turns, they may be tokenized differently: the assistant's generation produces splits conditioned on prior autoregressive context, while the user input is batch-encoded from a raw string. This produces a sequence where semantically identical words carry different token IDs depending on their speaker, potentially introducing inconsistencies in the attention patterns and gradient signal the model receives during training.

We propose a preprocessing method that addresses this inconsistency by aligning the tokenization of user input tokens to the splits observed in adjacent assistant tokens, using a vocabulary-constrained re-segmentation step. We hypothesize that this alignment will reduce training instability and improve convergence in multi-turn RL settings.

2 Problem Formulation

Let $\mathcal{C}^{(t)} = (s, u_1, a_1, u_2, a_2, \dots, u_t)$ denote a multi-turn conversation at turn t, where s is the system prompt, u_i is the i-th user message, and a_i is the i-th assistant response. During RL training, the model receives the full context $\mathcal{C}^{(t)}$ and is optimized to maximize expected reward on the generated response a_t.

The training sequence is constructed by tokenizing each component and concatenating the resulting token ID sequences:

$$\mathbf{x} = \tau(s) \oplus \tau(u_1) \oplus \mathbf{a}_1 \oplus \tau(u_2) \oplus \mathbf{a}_2 \oplus \cdots \oplus \tau(u_t) \quad (1)$$

where $\tau(\cdot)$ denotes batch tokenization from text and $\mathbf{a}_i$ denotes the token IDs produced during autoregressive generation of the i-th assistant turn.

The tokenization drift we identify arises when a lexical item w appears in both $\tau(u_i)$ and $\mathbf{a}_j$ for some i, j, but $\tau(w) \neq ids(w, \mathbf{a}_j)$, where $ids(w, \mathbf{a}_j)$ denotes the token ID subsequence corresponding to w within $\mathbf{a}_j$. This inconsistency is structurally inevitable when w does not correspond to a unique entry in the model vocabulary and its BPE segmentation depends on surrounding context.

3 Method: Context-Aware Tokenization Preprocessing

3.1 Overview

We propose a preprocessing step applied at the boundary between user input tokenization and the assembled training sequence. The key insight is that rather than tokenizing each user message independently, we tokenize it with awareness of the token ID vocabulary observed in the adjacent assistant turns. This produces user token IDs that are more likely to be consistent with the tokenization conventions of the surrounding assistant content.

3.2 Algorithm

Given the assistant token IDs $\mathbf{a}_i$ from the immediately preceding assistant turn, we construct a *local preferred split dictionary* $\mathcal{D}_i$ that maps surface-form substrings to their observed token ID sequences:

$$\mathcal{D}_i = \{(decode(\mathbf{a}_i[j:k]),\, \mathbf{a}_i[j:k]) \;:\; j < k \leq |\mathbf{a}_i|\} \tag{2}$$

We then tokenize the user message u_{i+1} using a constrained BPE procedure that, when a substring of u_{i+1} is present as a key in $\mathcal{D}_i$, preferentially applies the split observed in $\mathcal{D}_i$ rather than the default greedy merge. This is implemented as a postprocessing step over the standard tokenizer output: we scan the tokenized user sequence and identify subsequences whose decoded form matches an entry in $\mathcal{D}_i$ with a conflicting split, and replace them with the preferred split.

Formally, let $\hat{\tau}(u_{i+1}; \mathcal{D}_i)$ denote this context-aware tokenization. The aligned training sequence becomes:

$$\hat{\mathbf{x}} = \tau(s) \oplus \hat{\tau}(u_1; \mathcal{D}_0) \oplus \mathbf{a}_1 \oplus \hat{\tau}(u_2; \mathcal{D}_1) \oplus \mathbf{a}_2 \oplus \cdots \oplus \hat{\tau}(u_t; \mathcal{D}_{t-1})$$

where $\mathcal{D}_0$ is empty (no preceding assistant context at the first turn).

4 Experimental Setup

4.1 Models

We will conduct our experiments using Llama 3.2-1B (Grattafiori et al., 2024) and Qwen 3-0.6B (Yang et al., 2025).

Datasets: We will use **UltraChat** (Ding et al., 2023) and **LMRL-Gym** (Abdulhai et al., 2023) (tentative) for our experiments as they are both multiturn datasets.

We fine-tune each model using GRPO (Shao et al., 2024) in a multi-turn RL setting. All experiments use the same hyperparameters, reward model, and rollout procedure. The only variable is whether the proposed tokenization alignment preprocessing is applied to user turn tokens during training sequence construction.

5 Timeline

Honestly, this project might take a long while to complete, especially if we observe that the impact of tokenization drift is minimal in our chosen setup. The main bottleneck would be to validate how much impact this drift has on performance. Afterwards, all our experiments should be done within the space of 2 months.

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